
ECON 421 - INTRODUCTION TO GAME THEORY AND APPLICATIONS

博弈论导论及应用

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1 Introduction

Before we introduce what game theory is about, we first clarify what game theory is not about. Game theory is NOT about competitive markets.

Game theory evaluates strategic situations in which we analyze players' best courses of actions depending on opponents which lead to a Nash equilibrium or another form of equilibrium. Some classic examples are *prisoner's dilemma* and the *penalty kick*.

There are many forms of a game. This include *dynamic games* where players move at different times, *games of perfect information* which leads to *Zermelo's theorem*, *bargaining*, *signalling*, *reputation games*, and *auctions*.

In the following note, we go through them case by case where we give relevant definitions, analyses, and explanations.

2 Strategic Form and Normal Form

Strategic forms and normal forms are used to describe games where players *move simultaneously*. To describe such a game, we need the participating players, their available actions, and their corresponding payoffs in each outcome.

Definition 2.1. A strategic form is a list

$$G = (N; \{A_i\}_{i=1}^n; \{u_i\}_{i=1}^n)$$

where $N = \{1, 2, \dots, n\}$ represents the set of players, A_i is the set of available actions for player i , and u_i is the payoff function for player i with the mapping $A_1 \times A_2 \times \dots \times A_n \rightarrow \mathbb{R}$.

We assume that G is *common knowledge* to all players.

Definition 2.2. An action profile is given by a vector

$$\mathbf{a} = (a_1 \ a_2 \ \dots \ a_n)^\top$$

that is an element of $A := A_1 \times A_2 \times \dots \times A_n$.

We use the notation of “ $-i$ ” to denote the set excluding player i , that is player i ’s opponents decisions. For example, we can write an action profile as

$$\mathbf{a} = (a_i \ a_{-i})^\top$$

where

$$a_{-i} = (a_1 \ a_2 \ \dots \ a_{i-1} \ a_{i+1} \ \dots \ a_n)$$

Definition 2.3. A normal form is a matrix that describes the payoff of each player given their respective actions.

A 2-player game can be described as a single matrix, and a 3-player game can be described as two matrices. For example, the Prisoner’s Dilemma game can be described by the following matrix:

	Defect	Confess
Defect	(-1, -1)	(-3, 0)
Confess	(0, -3)	(-2, -2)

Definition 2.4. An action $a_i \in A_i$ is strictly dominated by a'_i if

$$u_i(a_i, a_{-i}) < u_i(a'_i, a_{-i})$$

for all $a_{-i} \in A_{-i}$.

A strictly dominated action should never be played by a rational player. If an action strictly dominates all other actions, it is a **strictly dominant action**.

Definition 2.5. An action $a_i \in A_i$ is **weakly dominated** by a'_i if

$$u_i(a_i, a_{-i}) \leq u_i(a'_i, a_{-i})$$

for all $a_{-i} \in A_{-i}$, **and**

$$u_i(a_i, a_{-i}) < u_i(a'_i, a_{-i})$$

for some $a_{-i} \in A_{-i}$

If an action weakly dominates all other actions, it is a **weakly dominant action**.

A **dominance solvable** game can be solved only by the players' dominant actions.

2.1 Second Price Auction

Consider n bidders, where each bidder can bid $b_i \geq 0$ who values the good at $v_i > 0$. The winner is the highest bidder but only pays the price of the second-highest bid. Therefore, the payoff for player i is

$$u_i(\mathbf{b}) = \begin{cases} v_i - \max_{j \neq i} b_j & b_i > \max_{j \neq i} b_j \\ \frac{1}{N(\mathbf{b})}(v_i - b_i) & b_i = \max_{j \neq i} b_j \\ 0 & b_i < \max_{j \neq i} b_j \end{cases}$$

where $N(\mathbf{b})$ is the number of tied winners.

Proposition 2.1. The best course of action for each player is to **bid their own value**, which is a **weakly dominant action for all players**.

We can show that $b_i = v_i$ is optimal, regardless of b_{-i} .

We first define the highest competing bid to be $\bar{b}_{-i} = \max_{j \neq i} b_j$, and divide the analysis into 3 cases.

1. Suppose $\bar{b}_{-i} > v_i$, in this case $u_i(v_i, \bar{b}_{-i}) = 0$.

1. If $b_i < \bar{b}_{-i}$, then $u_i(b_i, \bar{b}_{-i}) = 0$.
2. If $b_i = \bar{b}_{-i}$, then $u_i(b_i, \bar{b}_{-i}) = \frac{1}{N(\mathbf{b})}(v_i - b_i) < 0$.
3. If $b_i > \bar{b}_{-i}$, then $u_i(b_i, \bar{b}_{-i}) = v_i - b_i < 0$.

Hence, bidding v_i is optimal.

2. Suppose $\bar{b}_{-i} < v_i$, in this case $u_i(v_i, \bar{b}_{-i}) = v_i - \bar{b}_{-i} > 0$.

1. If $b_i < \bar{b}_{-i}$, then $u_i = 0$.
2. If $b_i = \bar{b}_{-i}$, then $u_i = \frac{1}{N(\mathbf{b})}(v_i - b_i) < v_i - \bar{b}_{-i}$.
3. If $b_i > \bar{b}_{-i}$, then $u_i = v_i - \bar{b}_{-i}$.

Hence, bidding v_i is optimal.

3. Suppose $v_i = \bar{b}_{-i}$, in this case $u_i(v_i, \bar{b}_{-i}) = \frac{1}{N(\mathbf{b})}(v_i - v_i) = 0$.

1. If $b_i < \bar{b}_{-i}$, then $u_i = 0$.
2. If $b_i = \bar{b}_{-i}$, then $u_i = \frac{1}{N(\mathbf{b})}(v_i - \bar{b}_{-i}) = 0$.
3. If $b_i > \bar{b}_{-i}$, then $u_i = v_i - \bar{b}_{-i} = 0$.

Hence, bidding v_i is optimal.

For all other actions $b'_i \neq v_i$, player i will either lose money if $v_i < \bar{b}_{-i} < b'_i$, or lose the auction if $b'_i < \bar{b}_{-i} < v_i$.

2.2 Iterated Elimination

This can be done based on the *common knowledge of rationality*, where all players are rational, and all players know their opponents are rational.

Additionally, no one should play strictly dominated actions and everyone knows other people will not do it.

Therefore, in the following example game:

	H	M	L
H	4, 4	0, 6	0, 5
M	6, 0	3, 3	1, 4
L	5, 0	4, 1	2, 2

We first eliminate the first row, because “H” is a strictly dominated action for player 1. Then we eliminate the first column from the rest, because it is a strictly dominated action for player 2. We do the same for the second row and the second column. This leaves us with (2, 2). This is a *Nash equilibrium*.

The concept of a Nash equilibrium lies within the best responses of players given “belief” (this will be discussed further) about their opponents’ actions. The notion is that players should *simultaneously* play their best-responses.

Definition 2.6. A (pure strategy) *Nash equilibrium* is an action profile

$$\mathbf{a}^* = \begin{pmatrix} a_1^* & a_2^* & \cdots & a_n^* \end{pmatrix}^\top$$

such that

$$u_i(a_i^*, a_{-i}^*) \geq u_i(a_i, a_{-i}^*)$$

for all $a_i \in A_i$.

The key of a Nash equilibrium is that no one wants to move away from it, as moving away from it will not be “profitable”.

In the Prisoner’s Dilemma game, the Nash equilibrium is (Confess, Confess), because if a player has a strictly dominant strategy, it must be played in a Nash equilibrium. Furthermore, if an action is weakly dominance for all players, the profile $\mathbf{a} = (a_i)_{i=1}^n$ is a Nash equilibrium (like the second price auction).

2.3 Cournot Competition Model

Let us think of a more general example to illustrate Nash equilibrium. Consider n firms, where each firm choose the amount to produce $q_i \geq 0$. The market price is determined by $P(Q) = 1 - Q = 1 - \sum_{i=1}^n q_i$. The marginal cost of production is $c_i \in (0, 1)$. The payoff for firm i is $u_i(q_i, q_{-i}) = q_i(P(Q) - c_i)$. To find the best-response of firm i , we maximize the payoff (profits):

$$\max_{q_i} u_i(q_i, q_{-i}) = q_i(1 - q_i - Q_{-i} - c_i)$$

where Q_{-i} is the total amount produced by its opponents.

This is a quadratic optimization problem, and the best-response only depends on total opponent production.

We already know that $u_i(0, q_{-i}) = 0$, $u_i(1, q_{-i}) < 0$, and $u_i''(q_i, q_{-i}) = -2 < 0$.

1. $u_i'(0) = 1 - Q_{-i} - c_i \leq 0$. This means $P(Q) - c_i = 1 - q_i - Q_{-i} - c_i < 0$. In this case, it is best for the firm to produce nothing. $BR_i(q_{-i}) = 0$.
2. $u_i'(0) = 1 - Q_{-i} - c_i > 0$, then we can maximize u_i by taking $u_i'(q_i) = 0$.

$$\begin{aligned} 1 - q_i - Q_{-i} - c_i - q_i &= 0 \\ q_i^* &= \frac{1 - Q_{-i} - c_i}{2} \end{aligned}$$

The best response is $BR_i(q_{-i}) = \frac{1 - Q_{-i} - c_i}{2}$.

Therefore, we summarize it to be:

$$BR_i(q_{-i}) = \begin{cases} \frac{1}{2}(1 - Q_{-i} - c_i) & 1 - Q_{-i} - c_i > 0 \\ 0 & 1 - Q_{-i} - c_i \leq 0 \end{cases}$$

If we take $c_i = c$ for all firms, the simultaneous best-responses leads to an equilibrium. Here we are also finding a *symmetric equilibrium* where identical players play the same action. This gives that:

$$q^* = \frac{1}{2}(1 - (n-1)q^* - c) \rightarrow q^* = \frac{1 - c}{n + 1}$$

and $P(Q) = 1 - nq^* = \frac{nc+1}{n+1}$.

As $n \rightarrow \infty$, $P(Q) \rightarrow c$.

2.4 Lobbying

Suppose there are l lobbyists, each of which makes a donation of $s_i \geq 0$, where the donation is spent regardless of the outcome. They only care about the their desired policy being enacted, and only one policy will be enacted. The value of the desired policy is $v > 0$, common to all lobbyists. The probability that lobbyist i wins is:

$$p_i(\mathbf{s}) = \frac{s_i}{\sum_{j=1}^l s_j}$$

The payoff for the lobbyist will be:

$$u_i(\mathbf{s}) = v \cdot p_i(\mathbf{s}) - s_i$$

and we define $p_i(s) = \frac{1}{l}$ if $s_1 = s_2 = \dots = s_l = 0$.

The game is symmetric, so we only need to consider symmetric strategy where $s_i^* = s^*$. Notice that all lobbyists choosing 0 is not a valid equilibrium, as any one of them making a donation will be profitable and they will deviate. Therefore, for each lobbyist, we need to maximize their payoff:

$$\max_{s_i} \frac{s_i}{s_i + s_{-i}} \cdot v - s_i$$

This means:

$$\frac{d}{ds_i} \frac{s_i}{s_i + s_{-i}} \cdot v - s_i = 0$$

which gives:

$$s_i = \sqrt{vs_{-i}} - s_{-i} \rightarrow s^* = \frac{v(l-1)}{l}$$

From here, we can calculate the total donation, and the equilibrium payoff for each lobbyist.

2.5 All-pay auction

Consider a different form of auction, where 2 bidders have value $v > 0$ for the good, and each places a bid of $b_i \geq 0$. Highest bidder wins the good (where ties are broken with $\frac{1}{2}$ chance for each side), but the this time, bids are paid regardless of winning or losing the auction. This means:

$$u_i(b_1, b_2) = \begin{cases} v - b_i & b_i > b_j \\ \frac{1}{2}v - b_i & b_i = b_j \\ -b_i & b_i < b_j \end{cases}$$

Notice that there is not a **pure strategy Nash equilibrium** here. This is where we introduce mixed strategies (the uncertainty about one's chosen action is modelled as a probability distribution).

2.6 Mixed Strategy

Let $\sigma_i \in \Delta(A_i)$, where $\sigma_i(a)$ represents the probability (density) of playing action a . This must satisfy that $\sigma_i(a) \in (0, 1)$ and $\sum_{a \in A_i} \sigma_i(a) = 1$ (or an integrates to 1 if the available actions are continuous). In this case, players make choices based on the expected payoff where:

$$U_i(\sigma) = \sum_{\mathbf{a} \in A} u_i(\mathbf{a})(\sigma_1(a_1) \cdots \sigma_n(a_n))$$

where σ is the mixed strategy profile.

Definition 2.7. A (mixed strategy) profile σ^* is a Nash equilibrium if

$$U_i(\sigma^*) \geq U_i(\sigma_i, \sigma_{-i}^*)$$

for all players and $\sigma_i \in \Delta(A_i)$

We can interpret the mixed strategy equilibrium in different ways:

1. Most frequent of different actions across multiple interactions.
2. Represents the beliefs about what opponents might do.
3. Population frequencies: the probability that represents shares of population that chooses certain action, or evolutionary game theory (frequency of action evolves based on how beneficial they are).
4. Represents unmodelled payoff uncertainty: purification – mixed strategy equilibrium is similar to pure strategy equilibrium in games with slight payoff perturbations.

There are some properties of a mixed strategy equilibrium that are useful.

1. Indifference: if actions a_i and a'_i are played with *positive probability*, then

$$u_i(a_i, \sigma_{-i}^*) = u_i(a'_i, \sigma_{-i}^*) = u_i(\sigma_i^*, \sigma_{-i}^*)$$

2. No Pure Deviation: it should also satisfy

$$u_i(a'_i, \sigma_{-i}^*) \geq u_i(a_i, \sigma_{-i}^*)$$

for all a'_i that are not used (played with probability 0).

These two properties are *equivalent to Nash optimality condition*, meaning that only when these two properties are satisfied, can there be a mixed strategy Nash equilibrium.

2.7 Crime Reporting

Consider N players observe a crime, and they can choose to report (R) or not report (N). The cost of reporting the crime is $c \in (0, 1)$ and the benefit from one report is $v > 1$. We won't talk too much about why there is not a pure strategy symmetric equilibrium, we discuss directly about mixed strategy equilibrium (there exists asymmetric pure strategy equilibrium, where only one person reports and no other people reports).

For each player, say the probability of reporting is $p \in (0, 1)$. The necessary condition is that:

$$\begin{aligned} U_i(R, \sigma_{-i}^*) &= U_i(N, \sigma_{-i}^*) \\ v - c &= v \cdot \mathbb{P}(\text{at least 1 player reports}) + 0 \cdot \mathbb{P}(\text{no player reports}) \\ v - c &= v \cdot (1 - (1 - p)^{N-1}) + 0 \cdot (1 - p)^{N-1} \\ p &= 1 - \left(\frac{c}{v}\right)^{\frac{1}{N-1}} \end{aligned}$$

The expected payoff for each player is $U_i(\sigma^*) = v - c$, which is inefficient, as it is equal to as if each individual is the only witness.

We can also analyze the effect of c , v and N on the probability that at least one player reports the crime, and we observe that as c increases, the probability decreases; as v increases, the probability increases; as N increases, the probability decreases, which leads to “free-riding problem”.

2.8 All-pay Auction (Mixed Strategy)

If we reconsider the all-pay auction through a mixed strategy lens, consider a symmetric mixed strategy such that the action of each player is chosen according to a CDF F and its corresponding PDF f . The bids $b_i \in [0, v]$ are all reasonable, where $b_i > v$ bids are strictly dominated. The expected payoff of player i when bidding b is:

$$\begin{aligned} U_i(b, F) &= (v - b)\mathbb{P}(b_j < b) + (-b)\mathbb{P}(b_j > b) + \left(\frac{v}{2} - b\right)\mathbb{P}(b_j = b) \\ &= vF(b) - bF(b) - b(1 - F(b)) \\ &= vF(b) - b \end{aligned}$$

Since the indifference property must be satisfied across all $[0, v]$, this means U_i must be constant for all $b \in [0, v]$:

$$\begin{aligned} U_i(0, F) &= U_i(v, F) = 0 \\ vF(b) &= b \\ F(b) &= \frac{b}{v} = \int_0^b f(x)dx \\ f(b) &= \frac{1}{v} \end{aligned}$$

Therefore,

$$F(b) = \begin{cases} 0 & b < 0 \\ \frac{b}{v} & b \in [0, v] \\ 1 & b > v \end{cases}$$

which gives

$$U_i(b, F) = \begin{cases} 0 & b \in [0, v] \\ v - b & b > v \end{cases}$$

The expected utility of each bidder is zero, and $(b_1, b_2) = (0, 0)$ is inefficient as Pareto dominates equilibrium.

The expected payment of each agent is:

$$\int_0^v bf(b)db = \frac{v}{2}$$

The total revenue from all-pay auction is v , where the total revenue is equal to the total effort. This is bad for participants, as gains from prize are spent as effort, but good for effort generation.

2.9 War of Attrition

The above section naturally leads to the War of Attrition game. Players decide how much time to wait, wait till time $t_i \geq 0$. The longest waiter wins, and the value of win is thus $v > 0$. The payoff is:

$$u_i(t_1, t_2) = \begin{cases} v - t_j & t_i > t_j \\ \frac{v}{2} - t_i & t_i = t_j \\ -t_i & t_i < t_j \end{cases}$$

The key thing here is that it is “combination” of all-pay auction and second price auction. You still need to wait, but only need to wait longer than your opponent.

The pure strategy equilibrium is either $(t_1 = 0, t_2 \geq v)$ or $(t_1 \geq v, t_2 = 0)$. This is because the best-response function is:

$$BR_i(t_j) = \begin{cases} (t_j, \infty) & t_j < v \\ \{0\} \cup (v, \infty) & t_j = v \\ \{0\} & t_j > v \end{cases}$$

In this case, we try to look for an equilibrium with symmetry, where players choose their action with CDF G and PDF g , where possible choices are $t \geq 0$.

Theorem 2.1. *Leibniz Rule* states that

$$\frac{d}{dt} \int_{\alpha(t)}^{\beta(t)} f(x, t) dx = \beta'(t) f(\beta(t), t) - \alpha'(t) f(\alpha(t), t) + \int_{\alpha(t)}^{\beta(t)} \frac{\partial}{\partial t} f(x, t) dx$$

Then, for player i , the expected payoff from choosing t is:

$$\begin{aligned} U_i(t, G) &= \int_0^\infty u_i(t, t_j) g(t_j) dt_j \\ &= \int_0^t (v - t_j) g(t_j) dt_j + \int_t^\infty (-t) g(t_j) dt_j \end{aligned}$$

The indifference condition states that $\frac{d}{dt} U_i(t, G) = 0$.

$$\begin{aligned} \frac{d}{dt} U_i(t, G) &= \frac{d}{dt} \left(\int_0^t (v - t_j) g(t_j) dt_j + \int_t^\infty (-t) g(t_j) dt_j \right) \\ &= \frac{d}{dt} \left(\int_0^t (v - t_j) g(t_j) dt_j - t(1 - G(t)) \right) \\ &= (v - t)g(t) - (1 - G(t) - tg(t)) = 0 \\ vg(t) &= 1 - G(t) \\ G(t) &= 1 - e^{-\frac{t}{v}} \end{aligned}$$

from $G(0) = 0$. This gives $g(t) = \frac{e^{-\frac{t}{v}}}{v}$.

3 Extensive Form

Extensive form is used to describe a multi-stage game.

Definition 3.1. An extensive form specifies a game tree, player functions, information sets, and payoffs.

The game tree is a set of nodes connected by directed branches where there exists a root node such that each node (final or non-final) has at most 1 incoming branch, and can be reached uniquely from the root. We denote the non-final nodes to be h .

The players is represented by a set $N := \{1, 2, \dots, n\}$, where each player has functions $P(h)$, and payoffs $u_i(h_f)$ (final nodes).

An information set is what cannot be distinguished by a player (a set of nodes). Within the same information set, it must be the same player to move, the same moves available. Every non-final node is in an information set, which can be singleton, and they form an information partition.

There are two types of multi-stage games. The game is either *perfect information* (players know all previous information, and information sets are all singletons) or *imperfect information* (at least one non-singleton information) set.

The strategy of a player represents what one should do in every information set (a complete contingent plan) even if some information sets are “unreachable”.

Definition 3.2. Let I_i be an information set for player i . For all I_i , $A(I_i)$ is the set of actions available at I_i . A strategy profile is $(s_1, \dots, s_n)$ where for all I_i , $s_i(I_i)$ is an action taken by i at I_i and we have $s_i(I_i) \in A(I_i)$.

We solve the game that ensures credible threats or behaviours (there can be other Nash equilibria) by looking for “smaller games” and determine what players would do in that situation.

Definition 3.3. A subgame is a subset of nodes such that it contains a specific $h \in H$ and contains all $\text{SUCC}(h)$. If $h \in I_i$ is in a subgame and $h' \in I_i$, then h' is also in the same subgame.

Definition 3.4. A subgame perfect equilibrium is a strategy profile that induces a Nash equilibrium in every subgame.

We find this kind of equilibrium through *backward induction*.

In finite perfect information games, all subgame perfect equilibria can be obtained by backward induction.

3.1 Perfect Information Games

Definition 3.5. Competitive games are a special class of zero-sum games, where outcomes are only victory, tie, or loss, with 2 players and finite moves.

Theorem 3.1. Zermelo's theorem states that in a competitive game, either player A has a strategy that guarantees “victory”, or player B has a strategy that guarantees “victory”, or both players have a strategy that guarantees “tie”.

This is shown through induction.

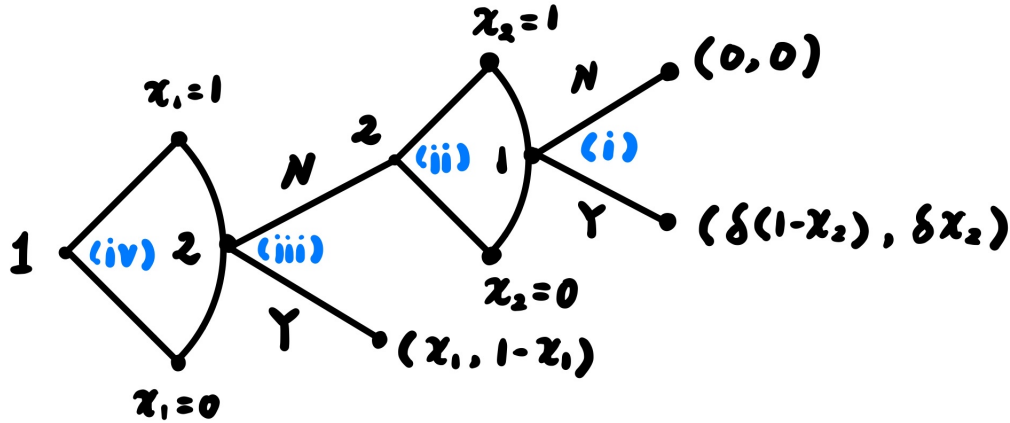
3.2 Bargaining

We consider a group of agents who need to divide a resource among themselves. A Rubinstein-Stahl models this by players making offers, which can be turned down.

A simple model is a single round of negotiation (**ultimatum game**) which consists of 2 players. Player 1 makes offer x_1 for itself, where $x_1 \in [0, 1]$ and player 2 decides Y or N. If player 2 choose Y, they get $(x_1, 1 - x_1)$; if player 2 choose N, they get $(0, 0)$.

In this case, the subgame perfect Nash equilibrium includes $r_2(x_1) = Y$, as it is always the best response of player 2 ($1 - x_1 \geq 0$). Since player 1 can choose any x_1 , it will choose x_1 that maximizes its profit. Thus, the SPE is $r_1(\cdot) = 1$, and $r_2(x_1) = Y$. Another best response by player 2 is to only say yes when player 1 offers something strictly less than 1, but in this case, there is no optimal strategy for player 1, so it is not part of the SPE.

Then, we consider a 2-round model, where player 2 has a chance to make an offer $x_2 \in [0, 1]$ if the first round of bargaining did not work. However, delay is costly. Thus, if agreement is reached in second round, the payoffs will be $(\delta(1 - x_2), \delta x_2)$, where $\delta \in (0, 1)$. The extensive form is:



The game at (ii) is an ultimatum game, which gives a payoff of $(0, \delta)$, where player 2 choose to offer $x_2 = 1$ and player 1 responds with Y.

We move one step back, we want to compare $1 - x_1$ and δ . Thus, at (iii), the best response of player 2 is:

$$r_2^*(x_1) = \begin{cases} Y & 1 - x_1 \geq \delta \rightarrow x_1 \leq 1 - \delta \\ N & 1 - x_1 < \delta \rightarrow x_1 > 1 - \delta \end{cases}$$

This implies the best response at (iv) by player 1 is $x_1 = 1 - \delta$, as it is the highest acceptable offer.

The SPE is then described to be: $r_1^*(\cdot) = 1 - \delta$, $r_2^*(x_1) = 1$, $r_1^*(x_1, N, x_2) = Y$.

Finally, we consider a “not-so-simple” model about this game. In real applications, players do not know the number of rounds or offers, there is uncertainty about time. This model uses an *infinite horizon* model, that is if an offer is rejected, another one is made next period, but delay is costly such that if the negotiation is too long, nothing is left. The discount rate for each player is δ_1 and δ_2 . The more impatient the player is, the smaller their δ would be. Note that in this case, a subgame at start of round n is identical to the subgame at start of round $n + 2$. Thus, for simplicity, we focus on stationary strategies, where players act similarly in similar situations, regardless of calendar date.

For player i , when offering, offers to keep x_i^* ; when accepting, will accept anything above the threshold of $\underline{x}_i$. Suppose no offers are accepted when $1 - x_1^* < \underline{x}_2$ and $1 - x_2^* < \underline{x}_1$. For the game to end somewhere, at least one of the inequality is violated.

In this case, in a certain round, assume player 1 is making an offer, and player 2 decides to accept the offer. This means $1 - x_1^* \geq \underline{x}_2$. To maximize profit, $x_1^* = 1 - \underline{x}_2$.

Then, we trace back one step, where player 2 is making an offer in the previous round, offering x_2 . If player 1 accepts, it gets $1 - x_2$; if it declines, it gets $\delta_1 x_1^*$ (from next round). This means player 1 will accept as long as $1 - x_2 \geq \delta_1 x_1^*$.

Now, we think about player 2 making this offer. If it offers such that $1 - x_2 \geq \underline{x}_1$, player 2 gets at most $x_2 = 1 - \underline{x}_1$; if it offers such that $1 - x_2 < \underline{x}_1$, player 1 will reject, and in the next round, player 2 will get $\delta_2(1 - x_1^*)$. However, it is easily shown that $1 - \underline{x}_1 \geq \delta_2(1 - x_1^*)$. This means $x_2^* = 1 - \underline{x}_1$.

By tracing one more step back, similar to the second step in analysis, we have $\underline{x}_2 = \delta_2 x_2^*$.

This gives us a system of equations:

$$\begin{cases} x_i^* = 1 - \underline{x}_j \\ \underline{x}_i = \delta_i x_i^* \end{cases}$$

for $i, j \in \{1, 2\}$. This solves to:

$$\begin{cases} x_i^* = \frac{1 - \delta_j}{1 - \delta_i \delta_j} \\ \underline{x}_i = \delta_i \cdot \frac{1 - \delta_j}{1 - \delta_i \delta_j} \end{cases}$$

If player 1 initiates this bargaining, the payoff is thus $(x_1^*, 1 - x_1^*)$ described above. By setting $\delta_i \approx 1$ and $\delta_i = 0$, we can obtain four situations of payoffs.

The role of patience is that say each stage lasts $\Delta > 0$, and players use interest rate $r_i > 0$ to indicate patience where $\delta_i = e^{-r_i \Delta}$. Fast offering, which is $\Delta \rightarrow 0$, will yield that:

$$x_1^* \rightarrow \frac{r_2}{r_1 + r_2}, \underline{x}_1 \rightarrow \frac{r_2}{r_1 + r_2}$$

4 Signalling

Signalling games involve choices of a party (sender) influencing the *beliefs*, and therefore, actions of other parties (receivers).

The sender knows its own characteristic, and makes a public decision (signal).

The receiver is interested in the underlying characteristic. It observes the signal (which it may or may not care).

The model is described in the following way. Consider a sender (1) and a receiver, where S observes type $\theta \in \Theta$ (we can assume the types are distributed with probability density $p(\theta) \in [0, 1]$) that is exogenous. S chooses an action $a_1 \in A_1$, and R observes a_1 to choose $a_2 \in A_2$. The payoff for player i is $u_i(a_1, a_2, \theta)$.

A pure strategy form (s_1, s_2) is the mapping where $s_1 : \Theta \rightarrow A_1$ and $s_2 : A_1 \rightarrow A_2$. A mixed strategy form is described to be $\sigma_1(a_1 | \theta) = \mathbb{P}(a_1 | \theta)$ and $\sigma_2(a_2 | a_1) = \mathbb{P}(a_2 | a_1)$. The beliefs is a function that $\mu : A_1 \rightarrow \Delta(\Theta)$. For example, $\mu(\theta | a_1)$ represents the probability assigned to type θ by the receiver once action a_1 has been observed.

Definition 4.1. A weak perfect Bayesian equilibrium satisfies two conditions:

1. Rationality condition: players choose optimal strategies given beliefs.
2. Consistency condition: beliefs are updated according to Bayes' rule.

A pure strategy wPBE is (s_1^*, s_2^*, μ) satisfying sequential rationality for S and R:

$$\forall \theta \in \Theta, s_1^*(\theta) \in \operatorname{argmax}_{a_1 \in A_1} u_1(a_1, s_2^*(a_1), \theta)$$

$$\forall a_1 \in A_1, s_2^*(a_1) \in \operatorname{argmax}_{a_2 \in A_2} \sum_{\theta} \mu^*(\theta | a_1) u_2(a_1, a_2, \theta)$$

and Bayes Consistency

$$\mu(\theta | a_1) = \begin{cases} \frac{\mathbb{P}(\theta)}{\mathbb{P}(a_1)} & s_1(\theta) = a_1 \\ 0 & s_1(\theta) \neq a_1 \end{cases}$$

where $\mathbb{P}(a_1) = \sum_{\{\theta' | s_1(\theta') = a_1\}} \mathbb{P}(\theta')$

4.1 Spencer's Education Model

Consider a sender to be a worker, and a receiver to be an employer. $A_1 = \mathbb{R}^+$ which indicates the education level, and $\Theta = \{L, H\}$ which indicates the productivity level. $A_2 = \mathbb{R}^+$ is the set of wages offered to the worker.

The payoffs are $u_1(a_1, a_2, \theta) = a_2 - \frac{a_1}{\theta}$ and $u_2(a_1, a_2, \theta) = -(a_2 - \theta)^2$.

There are three types of equilibrium: separating, pooling, and semi-pooling. A separating equilibrium is when the sender chooses distinct actions according to their types. This means their actions are fully revealing.

A pooling equilibrium is when the sender chooses the same action regardless of their type, which conveys no information.

For this specific model, we start by analyzing the sequential rationality condition of player 2.

$$\begin{aligned} s_2(a_1) &= \sum_{\theta} \mu(\theta \mid a_1) u_2(a_1, a_2, \theta) \\ &= \mu(L \mid a_1) (-(a_2 - L)^2) + \mu(H \mid a_1) (-(a_2 - H)^2) \end{aligned}$$

To maximize, we take the first-order condition, and we have:

$$\begin{aligned} \frac{d}{da_2} s_2 &= -2\mu(L \mid a_1)(a_2 - L) - 2\mu(H \mid a_1)(a_2 - H) = 0 \\ a_2^* &= \mu(L \mid a_1)L + \mu(H \mid a_1)H \end{aligned}$$

Then, we analyze the sequential rationality of player 1.

If it is a separating equilibrium, and player 1 plays (a_L^*, a_H^*) respectively when its type is (L, H) , then $s_2(a_L^*) = L$ and $s_2(a_H^*) = H$.

We want to make sure that when it is L type, it has no incentive to deviate, that is

$$u_1(a_L^*, L, L) \geq u_1(a_1, s_2^*(a_1), L)$$

when $a_1 = 0$, this gives that $L - \frac{a_L^*}{L} \geq s_2(0) - \frac{0}{L} \geq L$, which means $a_L^* = 0$.

We also want to ensure that if it is L type, it has no incentive to pretend it is H type

$$u_1(a_L^* = 0, L, L) \geq u_1(a_H^*, H, L)$$

and when it is H type, it has no incentive to pretend it is L type

$$u_1(a_L^* = 0, L, H) \leq u_1(a_H^*, H, H)$$

These two solve to $(H - L)L \leq a_H^* \leq (H - L)H$.

This means, a separating wPBE with the given choices exists if and only if $a_L^* = 0$ and a_H^* satisfies the derived condition.

It is already shown that if there exists a separating equilibrium, the above condition holds, however, we also need to show that if the condition holds, there exists a separating equilibrium. In this case, there are cases where the Bayesian beliefs cannot be updated (e.g. $a_1 \neq a_L^*, a_H^*$). We just say the employer assumes the worst and $\mu(H \mid a_1) = 0$ in these cases. If we write out all the beliefs and strategies of player 2, we can find the payoff for player 1 when it is L, is always 0; and when it is H, is $H - \frac{a_H^*}{H}$. The least-cost separating equilibrium is Pareto dominant, that is $a_H^* = (H - L)L$.

If it is a pooling equilibrium, that is $s_1^*(H) = s_1^*(L) = a^*$, then $\mu(H \mid a_1) = \mathbb{P}(H)$. The sequential rationality condition for player 2 is:

$$\bar{\theta} := \mathbb{P}(H)H + \mathbb{P}(L)L$$

Deviating to $a_1 = 0$ must not be profitable for player 1:

$$\bar{\theta} - \frac{a^*}{\theta} \geq s_2(0) - \frac{0}{\theta} \geq L$$

this means that $a^* \leq (\bar{\theta} - L)\theta$. A pooling equilibrium with signal a^* exists if and only if $a^* \leq (\bar{\theta} - L)L$. A complete equilibrium is described by $(s_1^*(\theta), s_2^*(a_1), \mu)$.

4.2 Intuitive Criterion

First [...] given the equilibrium outcome, a given out of equilibrium message is not sensible for certain type of [sender]. Second, insofar as this asserted fact is held by the players to be correct, introspection on their part will them that [receiver] will attach zero weight to this out-of-equilibrium message coming from those types [...]. A wPBE satisfies the intuitive criterion if the equilibrium belief μ^* is consistent with the intuitive logic. In the Spencer's model, this means three things:

1. If the intuitive criterion is satisfied, we must have

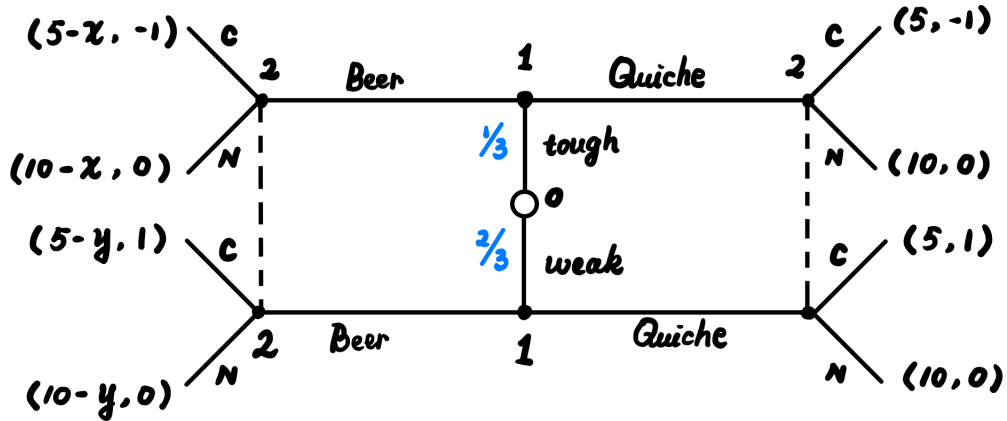
$$\mu(H | a_1) = 1, \forall a_1 \in (\underline{a}_H, a_H^*)$$

2. This cannot be the case in equilibrium since H types would want to deviate from a_H^* to a_1 , since employers would pay H to all education levels $a_1 \in (\underline{a}_H, a_H^*]$.
3. The only separating equilibrium satisfying the intuitive criterion is the least-cost separating equilibrium.

This is saying: if type L is clearly not attracted by education a , but type H potentially attracted, then a receiver would conclude that a has been chosen by type H. To be clearly not attracted by a , type L would not choose a even with the highest possible reward; to be potentially attracted by a , the highest possible reward would attract type H to choose a .

4.3 Beer-Quiche Game

Consider a game described as the following extensive form:



where $x, y > 0$ and C means challenge and N means not challenge.

We start by looking at player 2's sequential rationality:

1. If player 2 decides to challenge given $a_1 = K \in \{B, Q\}$, then

$$U_2(C | K) = \mu_K(-1) + (1 - \mu_K)(1) = 1 - 2\mu_K$$

where $\mu_K = \mu(T \mid K)$.

2. $U_2(N \mid K) = 0$.

In equilibrium, player 2's best response is:

$$a_T = \begin{cases} C & \mu_K < \frac{1}{2} \\ N & \mu_K > \frac{1}{2} \\ \text{indifferent} & \mu_K = \frac{1}{2} \end{cases}$$

Then, we look at player 1's sequential rationality.

First consider $a_T = a_W = B$, this gives $\mu_B = \frac{1}{3}$ which leads to $a_B = C$. Then $U_1(B \mid W) = 5 - y$ and $U_1(Q \mid W) = 5$ or 10. Choosing Q is better than B for weak type, so this cannot be a pooling equilibrium.

Then consider $a_T = a_W = Q$, this gives $\mu_Q = \frac{1}{3}$, meaning that $a_Q = C$. The most convenient belief is that $\mu_B = 0$ and $a_B = C$ as well. In this case,

$$U_1(Q \mid W) = 5, U_1(B \mid W) = 5 - y.$$

$$U_1(Q \mid T) = 5, U_1(B \mid T) = 5 - x.$$

This means, there is a pooling PBE in this case.

Furthermore, consider two types of separating PBE:

1. $a_T = B, a_W = Q$: this gives that $\mu_B = 1$, and $\mu_Q = 0$. Player 2 will play $a_B = N, a_Q = C$.

We need to have

$$U_1(B \mid T) \geq U_1(Q \mid T)$$

$$10 - x \geq 5$$

$$x \leq 5$$

$$U_1(B \mid W) \leq U_1(Q \mid W)$$

$$10 - y \leq 5$$

$$y \geq 5$$

2. $a_T = Q, a_W = B$: this gives that $\mu_B = 0, \mu_Q = 1$. Player 2 will play $a_B = C$ and $a_Q = N$.

We need to have

$$U_1(B \mid T) \leq U_1(Q \mid T)$$

$$5 - x \leq 10$$

$$U_1(B \mid W) \geq U_1(Q \mid W)$$

$$5 - y \geq 10$$

In this case, there cannot be a separating PBE. Only the first case stands.

Finally, we discuss a semi-pooling PBE, where we need mixed strategies. We define that

$$\sigma_\theta = \mathbb{P}(a_1 = B \mid \theta)$$

Then, we must have:

$$\begin{aligned}\mu_B = \mu(T \mid B) &= \frac{\frac{1}{3}\sigma_T}{\frac{1}{3}\sigma_T + \frac{2}{3}\sigma_W} = \frac{\sigma_T}{\sigma_T + 2\sigma_W} \\ \mu_Q = \mu(T \mid Q) &= \frac{\frac{1}{3}(1 - \sigma_T)}{\frac{1}{3}(1 - \sigma_T) + \frac{2}{3}(1 - \sigma_W)} = \frac{1 - \sigma_T}{(1 - \sigma_T) + 2(1 - \sigma_W)}\end{aligned}$$

We show that the semi-pooling equilibrium exists when $\sigma_T = 1$ and $\sigma_W \in (0, 1)$, and $\mu_B = \frac{1}{1+2\sigma_W}$, $\mu_Q = 0$.

Let $\alpha_K = \mathbb{P}(a_2 = C \mid a_1 = K)$ Then player 1 must have the indifference condition for weak type where

$$\begin{aligned}U_1(Q \mid W) &= U_1(B \mid W) \\ 5 &= (5 - y)\alpha_B + (10 - y)(1 - \alpha_B) \\ \alpha_B &= 1 - \frac{y}{5}\end{aligned}$$

However, this mix is optimal only if player 2 is indifferent from its sequential rationality:

$$\mu_B = \frac{1}{1 + 2\sigma_W} = \frac{1}{2} \rightarrow \sigma_W = \frac{1}{2}$$

We finally check that optimality of type T:

$$U_1(B \mid T) \geq U_1(Q \mid T) \rightarrow \alpha_B(5 - x) + (1 - \alpha_B)(10 - x) \geq 5$$

This gives that $y \geq x$.

5 Reputation

The setup for this game involves two players, where they play the game repeatedly. Player 1 has multiple possible types, they can have different payoff types, different strategies (commitment type: choose pre-specified action or strategy), or normal type, who maximizes some baseline payoff. Player 2 decides what to do today based on beliefs about player 1's type, and updates belief based on player 1's past actions. It also knows that player 1 has the incentive to exploit reputation.

5.1 Chain-store Game

Consider two players, the first player being a firm who chooses to sell high-quality good (H) or low-quality good (L), the second player being a consumer who chooses to buy (B) or not buy (N). For the firm, producing low-quality goods is more profitable, and consumer only wants to buy high-quality goods. The payoff matrix is thus:

	B	N
H	(2, 3)	(0, 2)
L	(3, 0)	(1, 1)

Notice that L is strictly dominant for the firm, so the only Nash equilibrium is (L, N) with payoff of (1, 1). So player 1 would like to convince player that it sells high-quality product.

Let's say this game is twice-repeated. Then in the second period, both players must play the static Nash equilibrium, meaning that regardless of first period, players anticipate (L, N) in the second period. This means the firm convincing the consumer that it sells high-quality good in first period is not profitable. So in the first period, both players will also choose (L, N), meaning there is no room for reputation.

However, let's consider a situation where with probability $\mu_0 \in (0, \frac{1}{4})$, player 1 always chooses H (it is commitment type), and with probability $1 - \mu_0$, it is normal type who maximizes payoff.

Denote that $\mu(H)$ to be the probability assigned to type C after H is observed, and similar for $\mu(L)$.

Now consider the action matrix:

	C	N	P2
$\emptyset$	H	(1)	(2)
HB	H	L	(3)
HN	H	L	(4)
LB	H	L	N
LN	H	L	N

If player 2 believes the probability of player 1 playing H is p in any period, the best response is computed by comparing the expected payoff from playing B vs playing N:

$$(3p + 0(1 - p)) - (2p + 1(1 - p)) = 2p - 1$$

This means,

$$s_2^*(a_1) = \begin{cases} B & p > \frac{1}{2} \\ N & p < \frac{1}{2} \\ \text{indifferent} & p = \frac{1}{2} \end{cases}$$

Examine the pure PBE with $s_1(\emptyset \mid N) = L$. This means that $\mu(H) = 1$, then $s_2(HB \text{ or } HN) = B$. In period 1, $\mathbb{P}(\text{P1 chooses H}) = \mu_0 < \frac{1}{2}$, $s_2(\emptyset) = N$. Then, examine type normal P1's rationality:

1. By playing L: payoff after two rounds is $1 + 1 = 2$ from $(L, N)_1, (L, N)_2$.
 2. By playing H: payoff after two rounds is $0 + 3 = 3$ from $(H, N)_1, (L, B)_2$.
- Deviating is profitable, this is not a PBE.

Examine the pure PBE with $s_1(\emptyset \mid N) = H$. This means that $\mu(H) = \mu_0$, then $s_2(HB \text{ or } HN) = N$. In period 1, $\mathbb{P}(\text{P1 chooses H}) = 0$ because P2 always chooses N, which means this is also not a PBE. This means there are no pure PBE.

Now consider a mixed strategy equilibrium where:

$$s_1(\emptyset \mid N) = \begin{cases} H & \sigma \\ L & 1 - \sigma \end{cases}, s_2(HB \text{ or } HN) = \begin{cases} B & \alpha \\ N & 1 - \alpha \end{cases}$$

We want to find σ and α and check all the equilibrium conditions hold. We first have:

$$\mu(C \mid H) = \frac{\mu_0}{\mu_0 + \sigma(1 - \mu_0)}$$

To find σ , we need P2 to be indifferent following $s_1(\emptyset) = H$.

$$\begin{aligned} U_2(s_2 = B) &= U_2(s_2 = N) \\ \mathbb{P}(a_{1,2} = H \mid a_{1,1} = H) &= \frac{1}{2} = \mu(H) \\ \mu_0 &= (1 - \mu_0)\sigma \\ \sigma &= \frac{\mu_0}{1 - \mu_0} \end{aligned}$$

To find α , we need P1 to be indifferent between H and L in period 1:

$$\begin{aligned} \text{expected total payoff, L in period 1} &= \text{expected total payoff, H in period 1} \\ 1 + 1 &= 0 + 3\alpha + 1(1 - \alpha) \\ \alpha &= \frac{1}{2} \end{aligned}$$

We check that it indeed is an equilibrium, that is it is optimal for P2 to choose N in first period:

$$\begin{aligned} \text{Payoff from N} - \text{Payoff from B} &= 2\mathbb{P}(\text{P1 chooses H}) + 1\mathbb{P}(\text{P1 chooses L}) \\ &\quad - [3\mathbb{P}(\text{P1 chooses H}) + 0\mathbb{P}(\text{P1 chooses L})] \\ &= 1 - 4\mu_0 > 0 \\ \mu_0 &< \frac{1}{4} \end{aligned}$$

The expected payoff can thus be calculated for P1 and P2.

6 Bayesian Games

Bayesian games contains a set of players, their available actions, payoffs, and additional to a strategic form, set of types for each player with their corresponding type distribution.

Definition 6.1. Let list $G := (I; p; \{A_i\}_{i=1}^n; \{u_i\}_{i=1}^n; \{\Theta_i\}_{i=1}^n)$ denote a Bayesian game, where

1. I is the set of players.
2. A_i is the set of available actions for player i .
3. u_i is the payoff function for player i .
4. Θ_i is the set of possible types for player i .
5. p_i is the probability density of types for player i .

The strategies in a Bayesian game is a map $s_i : \Theta_i \rightarrow A_i$ such that $s_i(\theta_i) \in A_i$, where it represents what action will be chosen when θ_i is observed. We focus on pure strategies, even though mixed strategies can also be considered.

Definition 6.2. A **Bayesian-Nash equilibrium** is a strategy profile $\mathbf{s}^* = (s_1^* \ s_2^* \ \dots \ s_n^*)^\top$ where $s_i(\theta_i)$ is optimal given $s_{-i}(\theta_{-i})$ and θ_i .

And $s_i^*(\theta_i)$ maximizes expected payoff

$$U_i(a_i, s_{-i} \mid \theta_i) = \mathbb{E}(u_i(a_i, s_{-i}(\theta_{-i}) \mid \theta_i))$$

Player i does not know opponents' types or actions.

6.1 Cournot Model

Consider a revised versoin of the Cournot model, where each firm has a cost c_i with distribution F_i over $[0, 1)$, and $q_i(c_i)$ maps cost to output. We should have:

$$q_i^*(c_i) = \operatorname{argmax}_{q_i \geq 0} (1 - q_i - q_j^*(c_j) - c_i)q_i$$

for all i . Since player i does not know about opponents' types or actions, we want the q_i that maximizes the expected value of the above expression. And we assume that we can do this linearly, then it becomes maximizing

$$(1 - q_i - \bar{q}_j - c_i)q_i$$

where we denote $\bar{q}_j = \mathbb{E}(q_j^*(c_j))$. Then, the best response of player i is:

$$q_i^*(c_i) = \begin{cases} \frac{1}{2}(1 - c_i - \bar{q}_j) & 1 - c_i - \bar{q}_j \geq 0 \\ 0 & 1 - c_i - \bar{q}_j < 0 \end{cases}$$

If we take the expected value of $q_i^*(c_i)$, we thus have:

$$\bar{q}_i = \frac{1}{2}(1 - \bar{c}_i - \bar{q}_j)$$

which gives a system of equations that solves $\bar{q}_1, \bar{q}_2$ that only depends on $\bar{c}_1, \bar{c}_2$, which can be substituted back to find q_i^* .

6.2 Auctions

6.2.1 First Price Auctions

Consider the type of each bidder to be $\theta_i \sim U(0, 1)$, with the payoff to be:

$$u_i(a_i, \theta) = \begin{cases} \theta_i - a_i & a_i > a_j \\ \frac{1}{2}(\theta_i - a_i) & a_i = a_j \\ 0 & a_i < a_j \end{cases}$$

We want to find a BNE in symmetric linear strategies where

$$b_i(\theta_i) = \alpha\theta_i + \beta_i$$

with $\alpha > 0, \beta \geq 0$.

The procedure is the following, assume j uses the strategy with α, β given, the optimal bidding for i must also use the strategy with the same α, β , and finally check that the strategy is indeed optimal.

Assume $b_2(\theta_2) = \alpha\theta_2 + \beta$, ($\tilde{\theta}$, means the variable θ is unknown to the current player)

$$\begin{aligned} U_1(b, \theta_1) &= (\theta_1 - b)\mathbb{P}(b_2(\tilde{\theta}_2) < b) + 0 \cdot \mathbb{P}(b_2(\tilde{\theta}_2) > b) + \frac{\theta_1 - b}{2}\mathbb{P}(b_2(\tilde{\theta}_2) = b) \\ &= (\theta_1 - b) \cdot F\left(\frac{b - \beta}{\alpha}\right) \end{aligned}$$

The probability distribution of a bid b follows that:

$$G(b) = \begin{cases} 1 & b > \alpha + \beta \\ \frac{b - \beta}{\alpha} & \beta \leq b \leq \alpha + \beta \\ 0 & b < \beta \end{cases}$$

We want to find α, β such that the optimal bid for P1 also follows the same strategy, meaning $\max_b U_1(b \mid \theta_1)$, that is:

$$\begin{aligned} \frac{d}{db} U_1 \Big|_{b=\alpha\theta_1+\beta} &= 0 \\ \frac{d}{db} (\theta_1 - b)G(b) &= -G(b) + (\theta_1 - b)g(b) = 0 \\ (\theta_1 - b) \cdot \frac{1}{\alpha} &= \frac{b - \beta}{\alpha} \\ b &= \frac{\theta_1 + \beta}{2} \end{aligned}$$

We thus have: $\alpha = \frac{1}{2}, \beta = 0$.

Check if this is optimal. We are trying to maximize:

$$U_i(b \mid \theta_i) = (\theta_i - b)G(b) = \begin{cases} (\theta_i - b)2b & b \in [0, \frac{1}{2}] \equiv \theta_i \in [0, 1] \\ \theta_i - b & b > \frac{1}{2} \end{cases}$$

The first case is quadratic and concave, so the (α, β) is indeed optimal for the strategy.

The expected revenue can be found with the following way. Revenue is always given by the highest bid: $b^{(1)}(\theta) = \max(b_1^*(\theta_1), b_2^*(\theta_2))$. The distribution of the highest bid is thus:

$$\begin{aligned} G^{(1)}(x) &= \mathbb{P}(b_1^*(\theta_1) \leq x \wedge b_2^*(\theta_2) \leq x) \\ &= \mathbb{P}(b_1^*(\theta_1) \leq x) \times \mathbb{P}(b_2^*(\theta_2) \leq x) \\ &= (G(x))^2 \\ &= 4x^2 \end{aligned}$$

which gives $g^{(1)}(x) = 8x$. The expected revenue is thus:

$$R^{FPA} = \int_0^{\frac{1}{2}} xg^{(1)}(x)dx = \frac{1}{3}$$

6.2.2 Second Price Auction

Contrary to the strategic form, we introduce a private value $\theta_i \geq 0$ to each bidder. We have previously shown that bid-your-own-value is optimal regardless of opponents' actions. In the case of a Bayesian game, bid-your-own-value remains to be a weakly dominant strategy. Thus, $b_i^*(\theta_i) = \theta_i$.

Now we consider its expected revenue. Let $\theta_i \sim U(0, 1)$. Then, the CDF of each bidder's bid is:

$$H(b) = \begin{cases} b & b \in [0, 1] \\ 1 & b > 1 \end{cases}$$

Then, $b^{(2)} = \min(b_1^*(\theta_1), b_2^*(\theta_2))$.

$$\begin{aligned}
H^{(2)}(b) &= \mathbb{P}(b^{(2)} \leq b) \\
&= 1 - \mathbb{P}(b^{(2)} > b) \\
&= 1 - \mathbb{P}(b_1 > b \wedge b_2 > b) \\
&= 1 - (1 - \mathbb{P}(b_1 \leq b))(1 - \mathbb{P}(b_2 \leq b)) \\
&= 1 - (1 - b)(1 - b) \\
&= 1 - (1 + b^2 - 2b) \\
&= 2b - b^2
\end{aligned}$$

This means: $h^{(2)}(b) = 2 - 2b$. Thus,

$$R^{SPA} = \int_0^1 b \cdot h^{(2)}(b) db = \frac{1}{3}$$

6.3 Revenue Equivalence

Consider two bidders, with values independently distributed on $[\underline{\theta}, \bar{\theta}]$ with CDF F .

Definition 6.3. An *auCTION format* specifies possible bids b_i , quantity function $q_i(b) \in [0, 1]$ and payment function $t_i(b) \geq 0$.

We fix an auction format q_i, t_i and fix the BNE of this auction to be b_i^* .

We define:

1. $q_i^*(\theta_1, \theta_2) = q_i^*(b_1^*(\theta_1), b_2^*(\theta_2))$.
2. $t_i^*(\theta_1, \theta_2) = q_i^*(b_1^*(\theta_1), b_2^*(\theta_2))$.
3. $\bar{q}_i(\theta_i) = \mathbb{E}(q_i^*(\theta_i, \tilde{\theta}_j))$ and $\bar{t}_i(\theta_i) = \mathbb{E}(t_i^*(\theta_i, \tilde{\theta}_j))$

We say the BNE is *efficient* if $q_i^* = 1$ whenever $\theta_i > \theta_j$. In an efficient equilibrium, $\bar{q}_i = \mathbb{P}(\theta_i > \tilde{\theta}_j) = F(\theta_i)$. The expected payoff of type θ_i is thus $U_i(\theta_i) = \theta_i \bar{q}_i - \bar{t}_i$.

Definition 6.4. A type θ_i could imitate θ'_i and bid $b_i^*(\theta'_i)$, obtaining $\theta_i \bar{q}_i(\theta'_i) - \bar{t}_i(\theta'_i)$. In equilibrium, the deviation is not profitable

$$\theta_i \in \operatorname{argmax}_{\theta'_i} \theta_i \bar{q}_i(\theta'_i) - \bar{t}_i(\theta'_i)$$

This is *incentive compatibility*.

The first-order condition is thus:

$$\begin{aligned}
\frac{\partial}{\partial \theta'_i} \theta_i \bar{q}_i(\theta'_i) - \bar{t}_i(\theta'_i) &= 0 \\
\theta_i \bar{q}'_i(\theta'_i) - \bar{t}'_i(\theta'_i) &= 0
\end{aligned}$$

for all θ'_i .

Using this, we differentiate U_i over θ_i and obtain:

$$\begin{aligned}\frac{d}{d\theta_i}U_i &= \bar{q}_i(\theta_i) + \theta_i \bar{q}'_i(\theta_i) - \bar{t}'_i(\theta_i) \\ &= \bar{q}_i(\theta_i)\end{aligned}$$

This means:

$$\begin{aligned}U_i(\theta_i) - U_i(\underline{\theta}) &= \int_{\underline{\theta}}^{\theta_i} U'_i(\theta'_i) d\theta'_i \\ U_i(\theta_i) &= U_i(\underline{\theta}) + \int_{\underline{\theta}}^{\theta_i} \theta_i \bar{q}_i(\theta'_i) d\theta'_i\end{aligned}$$

The lowest-type utility can be obtained from the following:

1. efficiency: $\bar{q}_i(\underline{\theta}) = F(\underline{\theta}) = 0$, which gives $U_i(\underline{\theta}) = -\bar{t}_i(\underline{\theta})$.
2. Bidder payment is $\bar{t}_i(\underline{\theta}) \geq 0$ and **voluntary participation** gives $U_i \geq 0$.

This means $U_i(\underline{\theta}) = -\bar{t}_i(\underline{\theta}) = 0$.

Proposition 6.1. Any 2 auction formats with BNE satisfying:

1. An efficient BNE exists, and
2. Voluntary participation

generates the same expected payoff $U_i(\theta_i)$ for each player and type.

The total surplus is defined to be the sum of payoff of player 1, payoff of player 2, and the revenue generated. If q_i^* is efficient, the total surplus is maximized: $TS(\theta) = \max(\theta_1, \theta_2)$. Any 2 efficient auctions generate the same expected total surplus. Therefore,

Theorem 6.1. Any 2 auction formats with BNE satisfying:

1. An efficient BNE exists, and
2. Lowest type has zero utility.

generate the same **expected revenue**.

This means, details of the auction format do not matter for expected revenue under efficiency. To increase revenue relative to a standard SPA, one must typically sacrifice efficiency (e.g. via reserve prices).

Consider a SPA with reserve price $r > 0$:

1. If all bids $< r$: good not sold (inefficient).
2. If exactly one bid $> r$: winner pays r .
3. If two or more bids $> r$: standard second-price rule.

For two bidders with $b_i \sim U(0, 1)$, the optimal reserve is:

$$r^* = \frac{1}{2}$$